

THE 5TH INTERNATIONAL CONFERENCE ON SMART GRID SYNCHRONIZED
MEASUREMENTS & ANALYTICS (SGSMA 2026)

Synchrophasor Anomaly Detection Competition Guide

Anomaly/Event Detection, Classification, and Localization
Using PMU Data on IEEE 39-Bus Test System

Masoud Barati¹

Payman Dehghanian²

David Celeita³

¹University of Pittsburgh

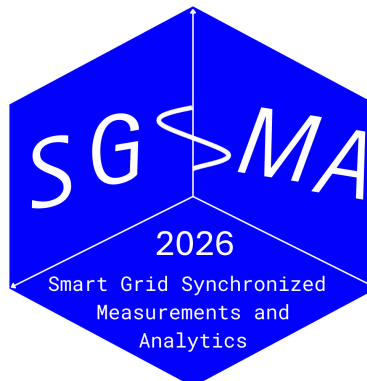
²George Washington University

³Universidad de la Sabana

📅 Competition Period: **February 15, 2026 to April 15, 2026**

📅 Announcement of the Top 3 High-Performing Teams: **April 25, 2026**

📅 On-site Competition Date (Top-3 Teams): **June 01, 2026** ⌚ Duration: **90-minute**



🎯 Competition Goal — Read This First

Design an algorithm that uses PMU time-series data to:

- 🔍 **Detect** abnormal behavior (normal vs. abnormal)
- 🔗 **Classify** the event type (fault, outage, generation trip, ...)
- 📍 **Localize** the event origin (bus and/or line)

while balancing **accuracy** and **model efficiency** (fewer parameters = bonus).

📋 Quick-Start Checklist

- 📄 **Input data:** 8 CSV files (one per PMU bus), timestamps aligned across all buses, ~162 000 rows each.
- 📁 **Your output:** Codes with predicted event labels & locations appended to files or in a separate submission CSV.
- 📄 **Report:** 3–6 page PDF with metrics, confusion matrix, parameter count, runtime, and training/validation curves.
- 🕒 **Data span:** ≈90 minutes at 30 frames/second (fps).

⚠️ Official Document

This guide is the authoritative reference for the SGSMA 2026 Synchrophasor Anomaly Detection Competition. If anything in supplementary files conflicts with this document, **this document takes precedence**. Contact the organizers at the first page of this document for clarifications.

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1 Introduction

1.1 What Are Phasor Measurement Units (PMUs)?

Phasor Measurement Units (PMUs) are high-precision sensors installed throughout the electric power grid. They provide **time-synchronized** measurements of voltage and current phasors at rates of 30–60 frames per second, all referenced to a common GPS clock with sub-microsecond accuracy.

By contrast, traditional Supervisory Control and Data Acquisition (SCADA) systems report data only every 2–10 seconds and without precise time synchronization. This makes SCADA blind to fast transients (lasting a few electrical cycles) and inter-area oscillations. PMUs fill this gap: their high sampling rate and precise time tags make it possible to observe **electromechanical dynamics** in real time.

What does a single PMU report?

At *every* time step ($\approx 1/30$ s), each PMU transmits the following:

- **3-phase voltage phasors** — magnitude (volts) and angle (degrees) for phases A, B, C.
- **3-phase current phasors** — magnitude (amps) and angle (degrees) for phases A, B, C.
- **Frequency** — system frequency in Hz (nominal: 60 Hz in North America).
- **ROCOF** — Rate of Change of Frequency (Hz/s); how fast frequency is changing.

A *phasor* captures both the **amplitude** of the AC signal (magnitude) and its **position in the 60 Hz cycle** (angle). Together these reveal power flow direction, voltage stability margins, and dynamic oscillations that are invisible to SCADA.

1.2 PMU Data as a Multivariate Time Series

PMU data is fundamentally a **multivariate time series**: an ordered sequence of feature vectors indexed by time. In this competition, each PMU generates one data frame every $\approx 1/30$ second, yielding a rich, high-dimensional stream:

- Each **row** is one time step (a single **TIMESTAMP**).
- Each **column** is a measurement feature (e.g., **VA_mag**, **Frequency**).
- **Patterns evolve** over time: ramps, oscillations, step changes, transient spikes — all carry event information.

Because consecutive samples are *strongly correlated*, standard machine-learning techniques that assume independent samples (e.g., random train/test splits) can lead to catastrophic data leakage. Time-series models — sliding windows, LSTMs, temporal CNNs, Transformers — are the natural tool here. Section 9 covers proper splitting in detail.

1.3 What Is Anomaly Detection on PMU Data?

Anomaly detection is the task of identifying when measurements deviate from “normal” operating behaviour. In PMU datasets, observed anomalies could fall into two broad groups:

1. **Physical events** — Faults (short circuits), transmission-line outages, generator trips, and sudden load changes that represent real disturbances on the grid.
2. **Data-quality issues** — Missing PMU frames (communication failures), corrupted measurements (bad data), and other instrumentation artefacts.

Rapid anomaly detection has direct operational value: faster situational awareness for control-room operators, reduced alarm fatigue, and enabling automated protective actions. In this competition, your algorithm must go *beyond* simple detection — it must also **classify** the anomaly type and **localize** where in the network it occurred.

2 Power System Model: IEEE 39-Bus Test Network

2.1 Overview

The competition dataset was generated from a dynamic time-domain simulation of the **IEEE 39-bus test system**, also known as the **10-machine New England system**. This benchmark model is one of the most widely used in power-systems research. Its key parameters are:

Component	Count	Parameter	Value
Buses	39	MVA base	100 MVA
Generators	10	Rated frequency	60 Hz
Transmission lines	34	HV voltage	345 kV
Transformers	12	Generator voltage	13.8 kV

2.2 One-Line Diagram with PMU Locations

Figure 1 shows the single-line diagram of the IEEE 39-bus test system. The **8 PMU locations** are marked with red labels (PMU 1 through PMU 8). Generators G1–G10 are connected via step-up transformers at buses 30–39. Understanding this topology is critical for the *localization* task: events propagate through the network, and the pattern of signatures across different PMUs encodes the event’s origin.

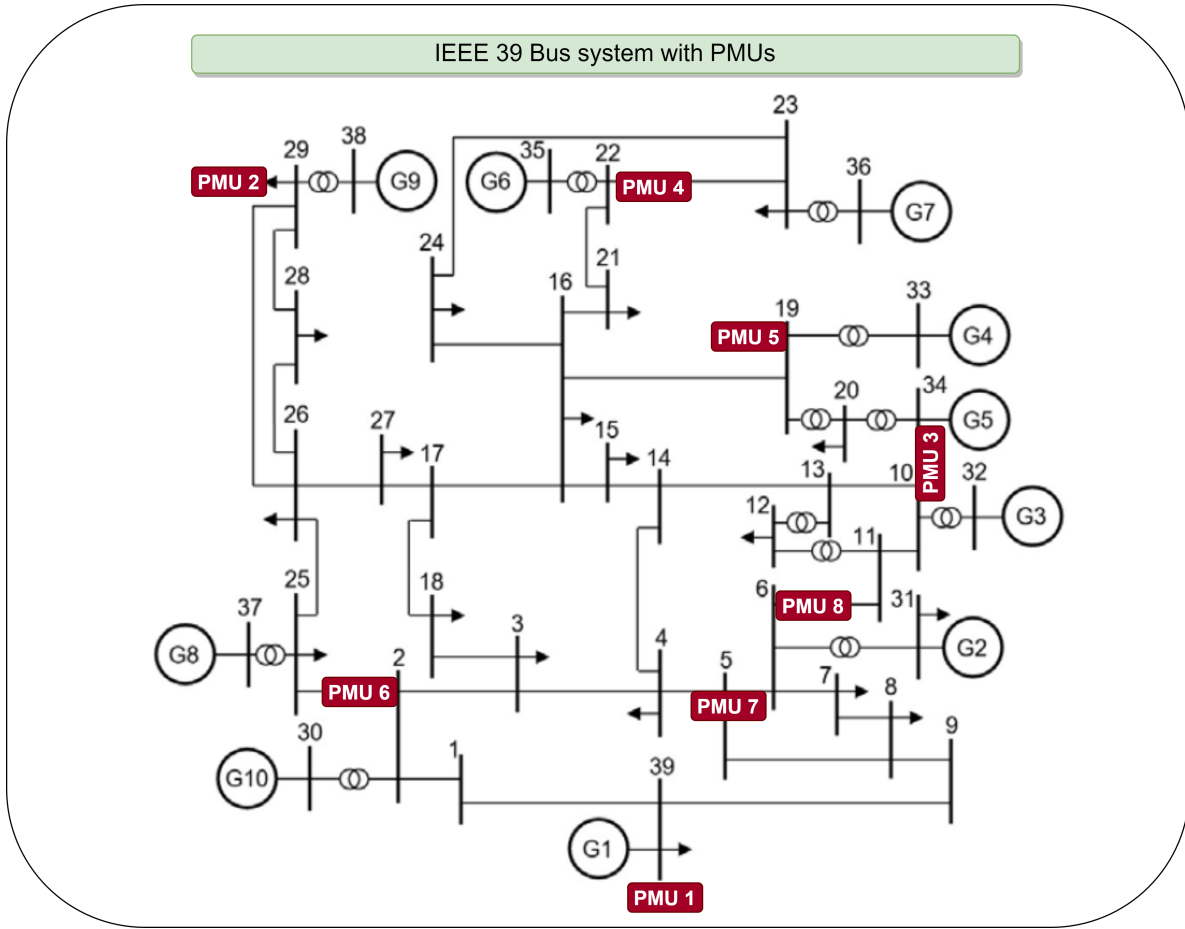


Figure 1: IEEE 39-bus (New England) test system one-line diagram. Red labels indicate the 8 PMU buses. Generators G1–G10 are at buses 30–39. The 345 kV transmission backbone interconnects load and generation buses.

2.3 PMU–Bus Mapping

Table 1 shows the exact mapping between PMU identifiers, bus numbers, base-case operating voltages from the power-flow solution, and the corresponding competition CSV filenames. This mapping is essential for interpreting your results and performing localization.

Table 1: PMU-to-bus mapping, base-case voltages, and data filenames (100 MVA base, 60 Hz).

PMU	Bus	kV	$ V $ (p.u.)	θ (deg)	CSV Filename
PMU 1	39	345.0	1.0300	−10.05	Bus39_Competition_Data_nanmask.csv
PMU 2	29	345.0	1.0499	+0.75	Bus29_Competition_Data_nanmask.csv
PMU 3	10	345.0	1.0172	−5.43	Bus10_Competition_Data_nanmask.csv
PMU 4	22	345.0	1.0498	+0.67	Bus22_Competition_Data_nanmask.csv
PMU 5	19	345.0	1.0499	−1.02	Bus19_Competition_Data_nanmask.csv
PMU 6	2	345.0	1.0487	−5.75	Bus2_Competition_Data_nanmask.csv
PMU 7	5	345.0	1.0053	−8.61	Bus5_Competition_Data_nanmask.csv
PMU 8	6	345.0	1.0077	−7.95	Bus6_Competition_Data_nanmask.csv

💡 Why does PMU placement matter?

Only 8 of the 39 buses have PMUs. When an event occurs at a *non-PMU* bus, your algorithm must infer its presence and location from **indirect** signatures observed at the 8 monitored buses. Events closer to a PMU produce stronger, faster signatures; distant events produce weaker, more diffuse responses.

This is why **multi-PMU fusion** and **knowledge of the network topology** are critical, especially for Task 3 (localization).

2.4 Complete Bus Data Reference

For participants who wish to compute electrical distances, build graph neural networks, or perform power-flow studies, Table 2 provides the full bus data for all 39 buses. The complete network model (including branch impedances, generator parameters, and loads) is available in the provided PSS/E file `IEEE_39_Bus_Power_System.raw`.

Table 2: Complete IEEE 39-bus test system steady-state data. PMU-equipped buses are **highlighted**. Bus types: 1 = PQ (load), 2 = PV (generator), 3 = Swing.

Bus	kV	Type	$ V $ (p.u.)	θ (deg)	PMU
BUS 1	345.0	PQ	1.0474	-8.44	—
BUS 2	345.0	PQ	1.0487	-5.75	PMU 6
BUS 3	345.0	PQ	1.0302	-8.60	—
BUS 4	345.0	PQ	1.0039	-9.60	—
BUS 5	345.0	PQ	1.0053	-8.61	PMU 7
BUS 6	345.0	PQ	1.0077	-7.95	PMU 8
BUS 7	345.0	PQ	0.9970	-10.12	—
BUS 8	345.0	PQ	0.9960	-10.61	—
BUS 9	345.0	PQ	1.0282	-10.32	—
BUS 10	345.0	PQ	1.0172	-5.43	PMU 3
BUS 11	345.0	PQ	1.0127	-6.28	—
BUS 12	345.0	PQ	1.0002	-6.24	—
BUS 13	345.0	PQ	1.0143	-6.10	—
BUS 14	345.0	PQ	1.0117	-7.65	—
BUS 15	345.0	PQ	1.0154	-7.73	—
BUS 16	345.0	PQ	1.0318	-6.19	—
BUS 17	345.0	PQ	1.0336	-7.30	—
BUS 18	345.0	PQ	1.0309	-8.22	—
BUS 19	345.0	PQ	1.0499	-1.02	PMU 5
BUS 20	110.0	PQ	0.9912	-2.01	—
BUS 21	345.0	PQ	1.0318	-3.78	—
BUS 22	345.0	PQ	1.0498	+0.67	PMU 4
BUS 23	345.0	PQ	1.0448	+0.47	—
BUS 24	345.0	PQ	1.0373	-6.07	—
BUS 25	345.0	PQ	1.0576	-4.36	—
BUS 26	345.0	PQ	1.0521	-5.52	—
BUS 27	345.0	PQ	1.0377	-7.49	—
BUS 28	345.0	PQ	1.0501	-2.01	—
BUS 29	345.0	PQ	1.0499	+0.75	PMU 2
BUS 30	13.8	PV	1.0475	-33.33	— (G10)
BUS 31	13.8	Swing	0.9820	-30.00	— (G2)
BUS 32	13.8	PV	0.9831	-27.43	— (G3)
BUS 33	13.8	PV	0.9972	-25.80	— (G4)
BUS 34	13.8	PV	1.0123	-26.82	— (G5)
BUS 35	13.8	PV	1.0493	-24.37	— (G6)
BUS 36	13.8	PV	1.0635	-21.68	— (G7)
BUS 37	13.8	PV	1.0278	-27.58	— (G8)
BUS 38	13.8	PV	1.0265	-22.19	— (G9)
BUS 39	345.0	PV	1.0300	-10.05	PMU 1 (G1)

3 Event Timeline and Ground-Truth Locations

The dataset contains a carefully orchestrated sequence of **physical** and **cyber** events injected into the simulation over the 90-minute window. Table 3 shows the complete event schedule. These ground-truth locations are provided so that participants understand the data structure; the per-sample **Event** label column in each CSV encodes these events at every time step.

Table 3: Event timeline for the competition dataset. Times are approximate; events may span several seconds around the listed time. “Cyber” events involve PMU communication failures; “Physical” events involve actual grid disturbances.

#	Category	Event Description	Approx. Time	Location
1	Cyber Event	Data drop at Bus 29 (PMU communication link failure — frames become NaN)	10th min	Bus 29
3	Physical Event	Three-phase line-to-ground (3LG) fault at Bus 39	20th min	Bus 39
4	Physical Event	Line outage between Bus 24 and Bus 23	40th min	Bus 24
5	Cyber Event	Data drop at Bus 29 (second communication failure)	45th min	Bus 29
7	Cyber + Physical	<i>Concurrent:</i> data drop at Bus 29 and generation change at Bus 2	50th min	Bus 29 & Bus 2
8	Physical Event	Generation change at Bus 2	50th min	Bus 2
9	Physical Event	Generation change at Bus 2	55th min	Bus 2
10	Physical Event	Load change at Bus 7	65th min	Bus 7
11	Physical Event	Load change at Bus 7	70th min	Bus 7

💡 Understanding the event timeline

- **Cyber events** (data drops) affect only the targeted PMU’s stream. During these intervals, the affected bus’s CSV will have `NaN` measurements and `DATA_PRESENT=0`. Other PMUs continue to report normally.
- **Physical events** (faults, outages, generation/load changes) produce electromechanical transients that propagate through the network. Their signatures appear at *multiple* PMUs with varying magnitude depending on electrical distance.
- **Event #7 is concurrent:** a cyber event at Bus 29 happens simultaneously with a physical generation change at Bus 2. Your algorithm must correctly handle this overlap — Bus 29’s data is missing while the generation event is happening, so detection depends on the *remaining* PMUs.
- **Note on numbering:** Event numbers #2 and #6 are not present in the table; the organizer numbering is non-consecutive. This does not affect the data — the **Event** column labels in the CSV are independent of these scenario numbers.
- **Spacing:** Some events are well-separated (>10 min apart) while others are closely spaced (5 min apart), testing your algorithm’s robustness under varying conditions.

4 Competition Package: Files Provided

The competition package contains the files listed below. Verify that you have **all** of them before you begin.

4.1 PMU Measurement Files (Required)

You will receive **8 bus-level CSV files**. Each contains time-aligned PMU measurements at 30 fps for ≈ 90 minutes ($\sim 162\,000$ rows per file, 17 columns each).

Filename	PMU	Bus
Bus2_Competition_Data_nanmask.csv	PMU 6	Bus 2
Bus5_Competition_Data_nanmask.csv	PMU 7	Bus 5
Bus6_Competition_Data_nanmask.csv	PMU 8	Bus 6
Bus10_Competition_Data_nanmask.csv	PMU 3	Bus 10
Bus19_Competition_Data_nanmask.csv	PMU 5	Bus 19
Bus22_Competition_Data_nanmask.csv	PMU 4	Bus 22
Bus29_Competition_Data_nanmask.csv	PMU 2	Bus 29
Bus39_Competition_Data_nanmask.csv	PMU 1	Bus 39

Using multiple PMU files together

Each CSV is one independent PMU data stream. You may:

- Build a **single-PMU model** that processes one bus at a time (simpler, but limited localization capability), **or**
- Build a **multi-PMU model** by merging all 8 files on the `TIMESTAMP` column (recommended for best performance, especially localization).

Because timestamps are aligned across all files, a simple `pandas.merge()` on `TIMESTAMP` produces a combined feature matrix with $8 \times 14 = 112$ PMU features plus 8 `DATA_PRESENT` flags. Multi-PMU models can exploit spatial correlations and differential responses to improve both classification and localization.

4.2 Power-System Metadata Files (Supplementary)

These supplementary files support topology-aware modelling, electrical-distance computation, and graph-based architectures:

File	Description
IEEE_39_Bus_Power_System.raw	PSS/E RAW format file with the full 39-bus network model (bus, branch, generator, and load data).
IEEE_39_Bus_Power_System_with_PMUs.png	One-line diagram with PMU locations (Figure 1).
PMUbus_Location.txt	Text file listing every bus, its voltage, power-flow angle, and PMU assignment.
Event_Timeline_&_Location.xlsx	Spreadsheet with event timeline (reproduced in Table 3).

Tooling note for the .raw file

The PSS/E RAW format requires a commercial licence to open in PSS/E. If you do not have access, use one of these free alternatives:

- **MATPOWER** (MATLAB/Octave) — [matpower.org](https://www.matpower.org)
- **pandapower** (Python) — pandapower.org (can import PSS/E RAW files directly)
- **PYPOWER** (Python) — github.com/rwl/PYPOWER
- **GridCal** (Python) — github.com/SanPen/GridCal

5 Dataset Format

Every CSV file has the **same column structure**. Each row represents one PMU measurement frame at a specific timestamp.

5.1 Column Description

Columns in each CSV file (17 columns total)

Column(s)	Description	#
TIMESTAMP	Elapsed time in seconds from simulation start, rounded to 3 decimals (≈ 30 fps spacing).	1
VA_mag, VA_ang VB_mag, VB_ang VC_mag, VC_ang	Three-phase voltage phasors: magnitude in volts and angle in degrees for phases A, B, C.	6
IA_mag, IA_ang IB_mag, IB_ang IC_mag, IC_ang	Three-phase current phasors: magnitude in amperes and angle in degrees for phases A, B, C.	6
Frequency	System frequency in hertz. Nominal = 60 Hz. Deviations indicate generation/load imbalance.	1
ROCOF	Rate of change of frequency (Hz/s). One of the fastest indicators of disturbances.	1
DATA_PRESENT	1 = PMU frame is valid; 0 = frame is missing (all measurements are NaN).	1
Event	Ground-truth label (integer 0–8). In the hidden test set, this column may be blank or withheld.	1

5.2 Understanding What Each Feature Tells You

Physical meaning of PMU features

- **Voltage magnitude** ($|V|$) reflects the voltage level at the bus. During faults, nearby buses experience sharp *voltage sags* (drops of 20–80%). During load changes, voltage drifts more gradually.
- **Voltage angle** (θ_V) indicates phase position relative to GPS time. Angle *differences* between buses reveal power flow direction and magnitude. During line outages, angles shift as power reroutes.
- **Current magnitude** ($|I|$) reflects the current flowing through the bus. Fault currents can spike to many times the normal value.
- **Current angle** (θ_I) changes with power factor and fault characteristics.

- **Frequency** (f) reflects the global generation–load balance. Loss of generation causes frequency to *drop*; loss of load causes it to *rise*. Normal deviations are <0.02 Hz; event deviations can reach 0.1–1.0 Hz.
- **ROCOF** (df/dt) is the time derivative of frequency. It responds *faster* than frequency itself and is often the earliest indicator of a disturbance. A sudden spike in ROCOF may precede observable changes in voltage or current by several hundred milliseconds.

6 Missing Data Representation

⚠ Rule for missing frames

When **Event = 5** (missing data only) or **Event = 6** (missing data concurrent with a physical event):

- **All 14 PMU measurement columns become NaN** (voltages, currents, frequency, ROCOF).
- **DATA_PRESENT = 0.**

This simulates a real-world PMU communication failure: the physical device is operational, but its data stream is interrupted and no measurements reach the control centre.

ℹ Strategies for handling NaN values

Choose one (or more) of the following approaches and clearly document your choice in the report:

1. **Masking:** Keep NaN values and supply a binary mask. The **DATA_PRESENT** column already serves this purpose. Models such as masked Transformers, attention mechanisms with padding masks, or graph neural networks with missing-node indicators handle this natively.
2. **Imputation:** Replace NaN values using:
 - Forward-fill / last-observation-carried-forward (LOCF),
 - Linear or spline interpolation, or
 - A learned imputation model (e.g., a variational autoencoder).

Important: Always keep **DATA_PRESENT** as an additional input feature, even after imputation. This lets the model learn that the filled values are synthetic.

3. **Hybrid:** Use imputation for classical ML pipelines (random forests, gradient boosting) and masking for deep sequence models.

💡 Practical advice on missing data

Missing-data periods are not random — they correspond to specific *cyber events* in the timeline (Table 3). During Event 6 (concurrent missing data + physical event), Bus 29’s PMU stream is unavailable *while* a generation change is happening at Bus 2. Your algorithm must detect the physical event from the **remaining 7 PMUs**. This scenario directly tests the robustness of your multi-PMU fusion approach.

7 Event Labels

The **Event** column uses integer labels **0–8**. Each label corresponds to a specific operating condition or disturbance type. A single integer is assigned per row (per PMU); concurrent events use label 6 or 8.

Table 4: Event label definitions.

ID	Name	Description and Physical Interpretation
0	Normal operation	No disturbance. The system is in steady state or near-steady state. Minor natural fluctuations may be present.
1	Fault	A short-circuit event (e.g., three-phase-to-ground). Produces sudden voltage sags and current spikes, typically lasting a few cycles to several seconds.
2	Line outage	A transmission line is disconnected (tripped or opened). Causes power-flow redistribution and voltage/angle shifts across the network.
3	Generation change/outage	A generator changes its MW output (step change) or trips offline entirely. Causes frequency deviation and system-wide power-flow redistribution.
4	Load change/-drop	A load suddenly increases, decreases, or disconnects. Similar to generation change but typically produces smaller frequency excursions.
5	Missing data	PMU frame missing due to communication failure . All measurements are NaN; <code>DATA_PRESENT=0</code> . No physical event is occurring.
6	Missing data + physical event	Missing data at one PMU concurrent with a physical event elsewhere. Measurements are NaN at the affected PMU; the physical event must be inferred from other PMUs.
7	Bad data	Corrupted measurement frame(s): non-physical spikes, jumps, or inconsistent values. The PMU reports data, but the values are unreliable.
8	Unknown event	An abnormal pattern that does not match labels 1–7. This is an <i>open-set</i> class for ambiguous or unusual disturbances.

⚠ Multi-event behaviour

If multiple physical events occur close together in time, overlapping transient patterns may appear across PMUs. In this dataset, concurrency is handled through:

- **Label 6** — when missing data at one PMU coincides with a physical event elsewhere.
- **Label 8** — when the combination of signatures does not cleanly match any single event class.

Your algorithm should be designed to handle these overlapping cases gracefully.

8 What You Are Expected to Build

Your solution should process streaming PMU data and produce three types of outputs. Each task builds upon the previous one.

🕒 Task 1: Event Detection (Binary Classification)

Objective: Determine whether the system is currently in **normal operation** (label 0) or **abnormal** (any of labels 1–8).

Key challenges:

- Low **false alarm rate** — avoid alerting during normal operation.
- Small **detection delay** — identify real events as quickly as possible after onset.
- The normal class dominates the dataset; your threshold must be carefully tuned.

Metrics: Precision, Recall, F1 for the Abnormal class; false alarm rate (FP/min); detection delay (seconds).

🕒 Task 2: Event Classification (Multi-Class Classification)

Objective: When an abnormal condition is detected, predict the specific **event type** (labels 1–8).

Key challenges:

- **Class imbalance** — faults (label 1) are brief and rare; normal operation (label 0) is abundant.
- Overlapping signatures — generation changes and load changes can look similar in frequency.
- Missing data (label 5) is easy to detect from NaNs; distinguishing label 5 from label 6 is harder.

Metrics: Macro-F1, Weighted-F1, per-class Precision/Recall/F1, full confusion matrix.

🕒 Task 3: Event Localization

Objective: Estimate the **bus and/or line** where the event originated.

Key challenges:

- Events at **non-PMU buses** must be inferred from indirect signatures.
- Multi-PMU fusion models that exploit the 39-bus topology outperform single-PMU approaches.
- Some events affect the same bus (Bus 2 has generation events at minutes 50 and 55; Bus 7 has load events at minutes 65 and 70); your model should correctly localize both.

Metrics: Top-1 accuracy, Top-3 accuracy, mean electrical-distance error (if provided).

8.1 Recommended Modelling Approaches

⚙️ Practical modelling guidance

- **Windowing:** Use sliding time windows of 1–5 seconds (30–150 samples) with an appropriate stride. Shorter windows yield faster detection; longer windows capture slower transients and oscillations.
- **Normalization:** Normalize each feature using *training-set* statistics. Robust scaling (median + IQR) is preferred because PMU data may contain extreme outliers during faults.
- **NaN handling:** Handle missing data explicitly (Section 6).

- **Multi-PMU fusion options:**

1. *Feature concatenation* — stack all 8 PMUs' features into a single wide vector ($8 \times 14 = 112$ features per time step).
2. *Attention-based fusion* — use cross-PMU attention to let the model learn which PMUs are most informative for each event.
3. *Graph neural network* — model the 39-bus topology as a graph, with PMU buses as observed nodes and line impedances as edge weights.

- **Model families to consider:**

- Temporal CNNs (1D convolutions over time)
- LSTM / GRU recurrent networks
- Transformer encoders (with positional encoding for time)
- Gradient-boosted trees (XGBoost, LightGBM) with hand-crafted window features
- Autoencoders (for unsupervised anomaly scoring)
- Hybrid pipelines (autoencoder for detection → classifier for typing)

9 Data Splitting and Leakage Prevention

PMU time series are **highly autocorrelated**: consecutive samples differ by tiny amounts. Randomly shuffling rows and then splitting into train/test sets leads to severe **data leakage** — the same event window (or near-identical samples) appears in both sets, inflating performance by 20–40% compared to realistic evaluations.

⚠ Do NOT randomly shuffle time-series rows!

Random shuffling of time-series data is the single most common methodological error in PMU anomaly-detection research. It produces artificially high accuracy that **does not generalize** to real-time deployment. Always use contiguous time-block splits.

⚠ Recommended split strategy

Unless the organizers provide a fixed train/test split, partition the data into **contiguous time blocks**:

- **70%** for training (earliest portion of the timeline)
- **15%** for validation (middle portion)
- **15%** for internal testing (latest portion)

Ensure that no sliding window straddles a split boundary. If necessary, leave a small buffer (e.g., one window-length) between splits.

i Why contiguous splits are essential

When you extract sliding windows from a time series, adjacent windows share most of their samples. If you shuffle and split at the row level, nearly identical windows end up in both training and test, making evaluation meaningless. Contiguous splits ensure that the model is evaluated on *future, unseen time periods*, which mirrors real-time deployment.

10 Evaluation and Reporting Metrics

To keep results **fair and comparable** across all teams, every submission **must** report the metrics below. The SGSMA organizers may additionally compute official scores on a hidden test set.

10.1 Task 1: Detection Metrics (Normal vs. Abnormal)

Binary classification where Normal = label 0 and Abnormal = labels 1–8.

- **Precision, Recall, F1-score** for the Abnormal class.
- **False alarm rate:** false positives per minute of normal operation.
- **Detection delay:** average time (in seconds) from event onset to first correct Abnormal prediction, if event boundaries are available.

10.2 Task 2: Classification Metrics (Multi-Class)

Multi-class classification over labels 0–8 (or 1–8 for abnormal-only scoring).

- **Macro-F1** — unweighted average of per-class F1 scores (treats all classes equally, including rare ones).
- **Weighted-F1** — per-class F1 weighted by support (reflects overall sample-level performance).
- **Per-class Precision, Recall, F1** — a table showing each class individually.
- **Confusion matrix** — full 9×9 matrix (labels 0–8).

10.3 Task 3: Localization Metrics

- **Top-1 accuracy:** fraction of events where the predicted location exactly matches ground truth.
- **Top-3 accuracy:** fraction where the true location is among the 3 most likely predictions.
- **Mean / median electrical-distance error** (if an electrical distance matrix is provided by the organizers).

10.4 Efficiency and Model Complexity

Efficient models are valued. Every team **must** report:

- **Trainable parameter count** (total number of learnable parameters).
- **Model size on disk** (MB), or a clear description for non-neural methods (e.g., number of trees $\times$ depth).
- **Inference time:** seconds required per minute of PMU data, and hardware used (CPU model, GPU model, RAM).
- **Window length:** number of time steps used for each detection/classification decision.

⚙️ Efficiency-aware scoring formula

The SGSMA evaluator team may use a composite score that rewards compact models:

$$\text{Score} = \text{Macro-F1} - \lambda \cdot \log_{10}(\text{Number of Parameters})$$

where λ is a small constant chosen by the evaluation committee.

Recommended range: $\lambda = 0.03$ as default; $\lambda \in [0.02, 0.05]$ as a sensible range.

Example: With $\lambda = 0.03$:

- A 100 K-parameter model pays a penalty of $0.03 \times 5 = 0.15$.
- A 10 M-parameter model pays $0.03 \times 7 = 0.21$.
- The 10 M model must have a Macro-F1 at least 0.06 higher to compensate for its size.

This encourages participants to pursue high accuracy *and* model simplicity.

11 Required Deliverables

Your final submission consists of **(A) Executable code with output prediction files** and **(B) a technical report**.

11.1 (A) Executable Code with Prediction Files

Submit your codes (in zipped file) with prediction files. **You should make sure that your code is executable on a standard machine by the competition organizing committee.** The predicted labels for the test data that can be included in one of the following formats:

1. **Inline:** Add columns `Predicted_Event` and (optionally) `Predicted_Location` to each bus CSV, aligned by `TIMESTAMP`.
2. **Separate file(s):** Provide CSV(s) with columns:
`TIMESTAMP`, `Bus`, `Predicted_Event`, `Predicted_Location`.

⚠️ Timestamp alignment is critical

If using separate prediction files, ensure `TIMESTAMP` values exactly match those in the provided data (same decimal rounding and row order). Misaligned timestamps will produce scoring errors or zero credit.

11.2 (B) Technical Report (PDF, 3–6 pages)

Your report should be concise, well-organized, and include **at minimum**:

1. **Method overview:** Model architecture, window length, features used, multi-PMU fusion strategy (if any).
2. **Preprocessing:** NaN handling (masking/imputation/hybrid), normalization, feature engineering.
3. **Training details:** Optimizer, learning rate, batch size, epochs, data augmentation, random seed(s).
4. **Results:**
 - Detection metrics (precision, recall, F1, false alarm rate).
 - Classification metrics (macro-F1, weighted-F1, per-class breakdown).

- Localization metrics (if applicable).
 - **Full confusion matrix.**
5. **Efficiency:** Parameter count, model size, inference time, hardware.
 6. **Training curves:** Loss and primary metric over epochs for both training and validation. Do *not* include test-set curves unless the organizers provide test labels.
 7. **Discussion:** Key design decisions, failure modes observed, and ideas for improvement.

12 Resources and Reference Models

12.1 IEEE 39-Bus Model Sources

Public repositories for the IEEE 39-bus system

- **Texas A&M Electric Grid Test Cases** (New England 39-bus): electricgrids.engr.tamu.edu/.../new-england-ieee-39-bus-system/
- **PSS/E RAW file downloads:** 3phaseee.com/raw_files

MATPOWER and case39

- MATPOWER official site: matpower.org
- Case39 reference: matpower.org/docs/.../case39.html
- Case39 source (GitHub): github.com/MATPOWER/.../case39.m
- MATPOWER User Manual: matpower.org/docs/MATPOWER-manual.pdf

12.2 Suggested Python Ecosystem

Recommended libraries

- **Data handling:** pandas, numpy, scipy
- **Machine learning:** scikit-learn, xgboost, lightgbm
- **Deep learning:** PyTorch or TensorFlow/Keras
- **Time-series:** tslearn, tsfresh (feature extraction), sktime
- **Power systems:** pandapower (for parsing the .raw file and computing electrical distances)
- **Visualization:** matplotlib, seaborn, plotly

Any programming language is acceptable, but Python is recommended for compatibility with modern ML frameworks.

13 Ethical Use and Academic Integrity

ⓘ Academic integrity and reproducibility

- Use the provided dataset **only** for the intended competition purposes.
- Do **not** attempt to infer or reverse-engineer ground-truth labels for any hidden test set.
- All code and writing must follow academic integrity standards, including proper attribution for external libraries, pre-trained models, and prior published work.
- Teams must ensure **reproducibility**: report random seeds, key hyperparameters, library versions, and enough implementation detail for organizers to verify the approach.
- Do **not** share competition data or solutions outside your team during the competition period.

14 Frequently Asked Questions

❗ Common questions answered

1. **Can I use pre-trained models?** Yes, but you must disclose all pre-trained components and count their parameters in the efficiency metric.
2. **Can I use external datasets for pre-training?** Using public IEEE 39-bus simulation data is generally acceptable. Using proprietary utility data is not. When in doubt, ask the organizers.
3. **What language should I use?** Any language (Python, MATLAB, Julia, R, etc.) is acceptable. Python is recommended for widest library support.
4. **Must I build all three tasks?** **Detection** and **Classification** are required. **Localization** is strongly encouraged and earns additional credit.
5. **How is the winner determined?** The SGSMA evaluation committee scores submissions on detection accuracy, classification performance (macro-F1), localization accuracy, model efficiency, and report quality. Exact weights will be announced by the organizers.
6. **What if my model cannot predict a certain class?** You may assign label 8 (unknown) when your model is uncertain. However, overusing label 8 will hurt your macro-F1 on the classes you are avoiding.
7. **Can I ensemble multiple models?** Yes. Report the *total* parameter count and inference time of the full ensemble.
8. **How do I compute electrical distance?** Parse the `.raw` file using `pandapower` or `MATPOWER` to obtain the bus admittance matrix Y_{bus} . Invert it to get the impedance matrix Z_{bus} ; the electrical distance between buses i and j is $|Z_{ii} + Z_{jj} - 2Z_{ij}|$.

Good Luck and Have Fun!

For questions, contact the SGSMA 2026 anomaly detection competition organizers.

✉ masoud.barati@pitt.edu

✉ payman@gwu.edu

✉ david.celeita@unisabana.edu.co